Theorem 1. *Let V be an integral operator on $L_2(\Gamma, w(x)dx)$ with integral kernel K satisfying*

$$|\partial_x^\alpha \partial_y^\beta K(x,y)| \leq c_{\alpha,\beta,K} \frac{|x-y|^{-|\alpha|_1-|\beta|_1}}{V(x,|x-y|)}.$$

We have

$$\|[M_f, V]\|_{\mathcal{L}_{N,\infty}(L_2(\Gamma,w(x)dx))} \leq c_K \|f\|_{\dot{W}^{1,N}(\Gamma)}.$$

Theorem 2. *Let $1 \neq g \in G$. Let V be an integral operator on $L_2(\Gamma, w(x)dx)$ with integral kernel K satisfying*

$$|\partial_x^\alpha \partial_y^\beta K(x,y)| \leq c_{\alpha,\beta,K} \frac{|x-y|^{1-|\alpha|_1-|\beta|_1}}{|x-gy| \cdot V(x,|x-y|)}.$$

We have

$$\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Gamma,w(x)dx))} \leq c_K \|f\|_{\dot{W}^{1,N}(\Gamma)}.$$

The first one should follow from Hytonen. The second is already written by Zhijie (though the proof is scattered across sections and is poor).

The tasks are (a) to write the proof of Theorem 1 modulo Hytonen (b) to write the proof of Theorem 2 in a clear, comprehensible way (c) to infer the upper estimate from Theorems 1 and 2.